

Tutorial - Week 12

TG 11 - TG 12

1. (a) Prove that for any convergent sequence $\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n$.
 (b) Consider a_n given by $a_1 = 1$, $a_{n+1} = \frac{1}{1 + a_n}$ for n in $\mathbb{N}$ and assume that $\{a_n\}$ is convergent. Find $\lim_{n \rightarrow \infty} a_n$.
2. (a) Show that if $\lim_{n \rightarrow \infty} a_{2n} = L$ and $\lim_{n \rightarrow \infty} a_{2n+1} = L$, then $\{a_n\}$ is convergent and $\lim_{n \rightarrow \infty} a_n = L$.
 (b) Define $a_1 = 1$ and $a_{n+1} = 1 + \frac{1}{1 + a_n}$ for n in $\mathbb{N}$. Find the first eight terms of the sequence $\{a_n\}$, and show that

$$\lim_{n \rightarrow \infty} a_n = \sqrt{2}.$$

This gives the *continued fraction expansion*

$$\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \dots}}$$

3. Determine whether each of the following series is absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{5 + 3n}$

(b) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n} + 1}$

(c) $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^2 + n - 1}$

(d) $\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{2n + 3}$

(e) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 + 4}$

(f) $\sum_{n=1}^{\infty} (-1)^{n+1} n e^{-n}$

(g) $\sum_{n=1}^{\infty} (-1)^{n-1} e^{2/n}$

(h) $\sum_{n=0}^{\infty} \frac{\sin\left((n + \frac{1}{2})\pi\right)}{1 + \sqrt{n}}$

(i) $\sum_{n=1}^{\infty} \frac{n \cos n\pi}{2^n}$

(j) $\sum_{n=1}^{\infty} (-1)^n \sin\left(\frac{\pi}{n}\right)$

(k) $\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{\pi}{n}\right)$

(l) $\sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$

(m) $\sum_{n=4}^{\infty} \frac{1}{2 + 3^n}$

(n) $\sum_{n=0}^{\infty} \frac{2^n}{n!}$

(o) $\sum_{n=1}^{\infty} n e^{-n^2}$

(p) $\sum_{n=1}^{\infty} \frac{e^{\sqrt{n}}}{n!}$

(q) $\sum_{n=2}^{\infty} \frac{\sqrt{3}^n}{3^{2n}}$

(r) $\sum_{n=1}^{\infty} \left(\frac{8n^2 + 1}{5n^2 - 1}\right)^{\frac{1}{2}}$

(s) $\sum_{n=0}^{\infty} (-1)^n \left(\frac{8n^2 + 1}{5n^2 - 1}\right)^{\frac{1}{2}}$

(t) $\sum_{n=1}^{\infty} \left(\frac{3n + 1}{8n^5 - 7}\right)^{\frac{1}{2}}$

$$(u) \sum_{n=1}^{\infty} (-1)^n \left(\frac{3n+1}{8n^5-7} \right)^{\frac{1}{2}}$$

$$(x) \sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{(\ln n)^n}$$

$$(v) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$$

$$(y) \sum_{n=0}^{\infty} (\arctan n)^n$$

$$(w) \sum_{n=1}^{\infty} (-1)^n \frac{2^n n!}{5 \cdot 8 \cdot 11 \cdots (3n+2)}$$

$$(z) \sum_{n=1}^{\infty} \left(1 - \frac{2}{n} \right)^{n^2}$$

4. (a) Analyze the convergence of the series $\sum_{n=1}^{\infty} (-1)^n \frac{n!}{n^n}$.
- (b) Approximate the series $\sum_{n=1}^{\infty} (-1)^n \frac{n!}{n^n}$ by using the sum of the first 10 terms. Estimate the error involved in this approximation.
- (c) How many terms are required to ensure that the partial sum S_n approximates the series within 10^{-4} ?

5. A sequence $\{a_n\}_n$ is defined by the equations

$$a_1 = 1, \quad a_{n+1} = \frac{2 + \cos n}{\sqrt{n}} a_n$$

Determine whether $\sum a_n$ converges or diverges.